CSE113
Foundations of Computer Science

State University of New York, Korea

Set

"a language in which we formulate and discuss the basic notions concerning collections of objects."

Roster Notation

- A = {1,2,3,4,5}
- Even = { 2, 4, 6, 8…}
- Prime = { 2, 3, 5, 7, ….}

Set-builder notation

Set builder notation describes a set that is defined by a predicate:
- {x ∣ P(x)}
- {x : P(x)}

Where the vertical bar (or colon) is a separator that can be read as "such that", "for which", or "with the property that".

A domain E can appear on the left of the vertical bar

$$\{x \in E \mid \Phi(x)\},$$

or by adjoining it to the predicate:

$$\{x \mid x \in E \text{ and } \Phi(x)\} \quad \text{or} \quad \{x \mid x \in E \wedge \Phi(x)\}.$$

Set-builder notation and roster notation

1. $\{n : n \text{ is a prime number}\} = \{2, 3, 5, 7, 11, 13, 17, \ldots\}$
2. $\{n \in \mathbb{N} : n \text{ is prime}\} = \{2, 3, 5, 7, 11, 13, 17, \ldots\}$
3. $\{n^2 : n \in \mathbb{Z}\} = \{0, 1, 4, 9, 16, 25, \ldots\}$
4. $\{x \in \mathbb{R} : x^2 - 2 = 0\} = \{\sqrt{2}, -\sqrt{2}\}$
5. $\{x \in \mathbb{Z} : x^2 - 2 = 0\} = \emptyset$
6. $\{x \in \mathbb{Z} : |x| < 4\} = \{-3, -2, -1, 0, 1, 2, 3\}$
7. $\{2x : x \in \mathbb{Z}, |x| < 4\} = \{-6, -4, -2, 0, 2, 4, 6\}$
8. $\{x \in \mathbb{Z} : |2x| < 4\} = \{-1, 0, 1\}$

Subsets

Definitions

- Subset. $A \subseteq B \Leftrightarrow \forall x,$ if $x \in A$ then $x \in B$
- Not subset. $A \not\subseteq B \Leftrightarrow \exists x$ such that $x \in A$ and $x \notin B$

Exercises

Let $A = \{\{0, 10\}, \{10, 100\}, 0\}$. Determine if the following are true or false.

1. $0 \in A$
2. $\{10\} \in A$
3. $\{10\} \subseteq A$
4. $\{0, 10\} \subseteq A$

Set equality

Definition

- Given sets A and B, A equals B, written $A = B$, if, and only if, every element of A is in B and every element of B is in A.
- $A = B \Leftrightarrow A \subseteq B$ and $B \subseteq A$.

Exercises

- $A = \{m \in \mathbb{Z} \mid m = 2a \text{ for some integer } a\}$
  $B = \{n \in \mathbb{Z} \mid n = 2b - 2 \text{ for some integer } b\}$
  Is $A = B$?

Exercises

Let Z denote the set of integers.

Let A = {7a+3b | a, b \in Z }

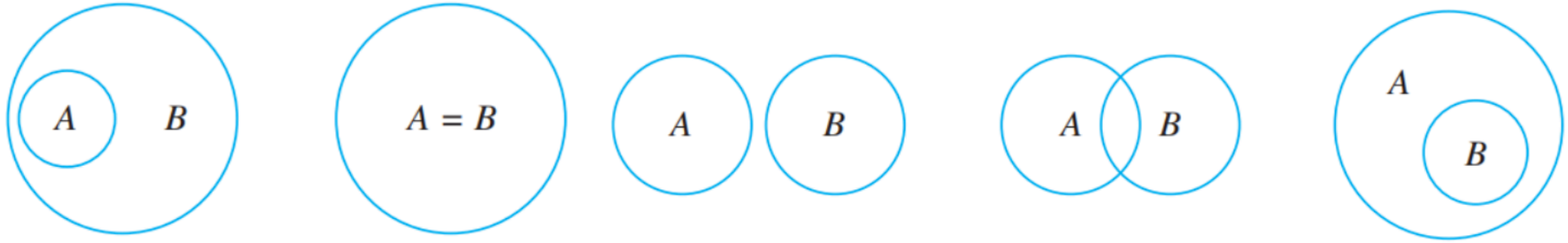
True or False: A = Z ?

Solution: This set contains all numbers of form $7a + 3b$, where a and b are integers. Each such number $7a + 3b$ is an integer, so A contains only integers. But *which* integers? If n is *any* integer, then $n = 7n + 3(-2n)$, so $n = 7a + 3b$ where $a = n$ and $b = -2n$. Thus $n \in A$, and so $A = \mathbb{Z}$. ✍

Venn diagrams

Definition

- Relationship between a small number of sets can be represented by pictures called Venn diagrams



(a) (b) (a) (b) (c)

Exercises

- Draw a Venne diagram representing
  - N: the set of natural numbers
  - Z: the set of integers
  - Q: The set of rationals
  - R: the set of reals

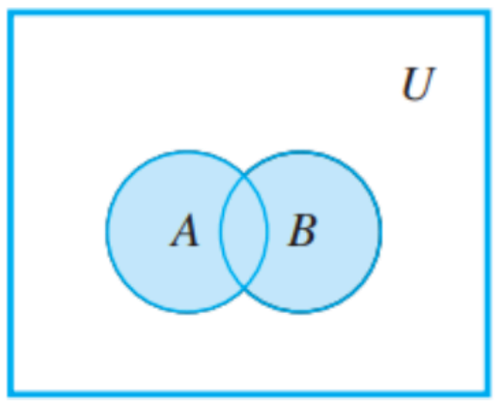
Operations on sets

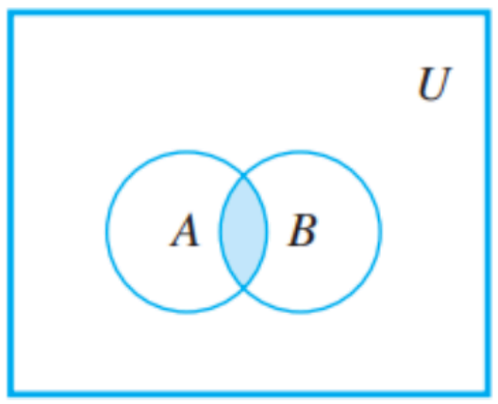
Definition

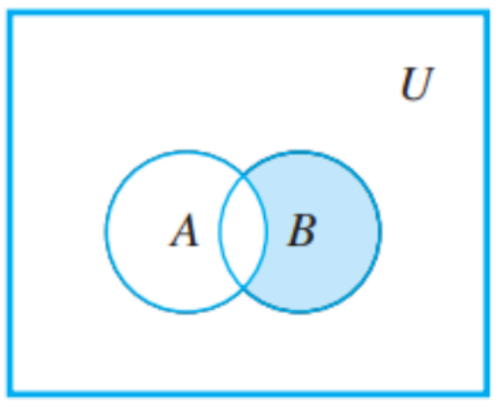
Let A and B be subsets of a universal set U.

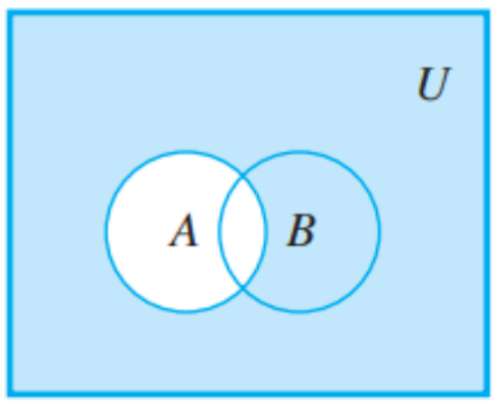
1. The union of A and B, denoted $A \cup B$, is the set of all elements that are in at least one of A or B.
   $A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$
2. The intersection of A and B, denoted $A \cap B$, is the set of all elements that are common to both A and B.
   $A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$
3. The difference of B minus A (or relative complement of A in B), denoted $B - A$, is the set of all elements that are in B and not A.
   $B - A = \{x \in U \mid x \in B \text{ and } x \notin A\}$
4. The complement of A, denoted A', is the set of all elements in U that are not in A.
   $A' = \{x \in U \mid x \notin A\}$

Operations on sets









Problems

- Let the universal set $U = \{a, b, c, d, e, f, g\}$.
  Let $A = \{a, c, e, g\}$ and $B = \{d, e, f, g\}$.
  Find $A \cup B$, $A \cap B$, $B - A$, and A'.

Notations (1)

- Closed interval: $[a,b] = \{x \in \mathbb{R} : a \le x \le b\}$



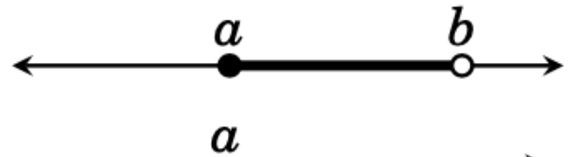
- Open interval: $(a,b) = \{x \in \mathbb{R} : a < x < b\}$



- Half-open interval: $(a,b] = \{x \in \mathbb{R} : a < x \le b\}$



- Half-open interval: $[a,b) = \{x \in \mathbb{R} : a \le x < b\}$



- Infinite interval: $(a,\infty) = \{x \in \mathbb{R} : a < x\}$



- Infinite interval: $[a,\infty) = \{x \in \mathbb{R} : a \le x\}$



- Infinite interval: $(-\infty,b) = \{x \in \mathbb{R} : x < b\}$



- Infinite interval: $(-\infty,b] = \{x \in \mathbb{R} : x \le b\}$



Exercises

- $A = (-1, 0] = \{x \in \mathbb{R} \mid \ -1 < x \leq 0\}$
  $B = [0, 1) = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$.
  Find $A \cup B$, $A \cap B$, $B - A$, and A'.

Notations (2)

Given sets $A_0, A_1, A_2, \ldots$ that are subsets of a universal set U and given a nonnegative integer n,

- $\cup_{i=0}^{n} A_i = \{x \in U \mid x \in A_i \text{ for at least one } i = 0, 1, 2, \ldots, n\}$
- $\cup_{i=0}^{\infty} A_i = \{x \in U \mid x \in A_i \text{ for at least one whole number } i\}$
- $\cap_{i=0}^{n} A_i = \{x \in U \mid x \in A_i \text{ for all } i = 0, 1, 2, \ldots, n\}$
- $\cap_{i=0}^{\infty} A_i = \{x \in U \mid x \in A_i \text{ for all whole numbers } i\}$

Exercises

- For each positive integer i, let
  $A_i = \{x \in \mathbb{R} \mid -\frac{1}{i} < x < \frac{1}{i}\} = \left(-\frac{1}{i}, \frac{1}{i}\right)$
  Find $A_1 \cup A_2 \cup A_3$ and $A_1 \cap A_2 \cap A_3$
  Find $\cup_{i=0}^{\infty} A_i$ and $\cap_{i=0}^{\infty} A_i$

Empty set

Definition

Empty set, denoted by ϕ, is a set with no elements.

Examples

- $\{1, 3\} \cap \{2, 4\} = \phi$
- $\{x \in \mathbb{R} \mid x^2 = -1\} = \phi$
- $\{x \in \mathbb{R} \mid 3 < x < 2\} = \phi$

Disjoint sets

Definition

- Two sets are called disjoint if, and only if, they have no elements in common.
- A and B are disjoint $\Leftrightarrow A \cap B = \phi$

Problems

- Let $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$. Are A and B disjoint?

Power set

Definition

- Given a set A, the power set of A, denoted $P(A)$, is the set of all subsets of A.

Problems

- Find the power set of the set $\{x, y\}$. That is, find $P(\{x, y\})$.

Cartesian product

Definition

- Given sets $A_1, A_2, \ldots, A_n$, the Cartesian product of $A_1, A_2, \ldots, A_n$ denoted $A_1 \times A_2 \times \cdots \times A_n$, is the set of all ordered n-tuples $(a_1, a_2, \ldots, a_n)$ where $a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n$.
- $A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n\}$.

Problems

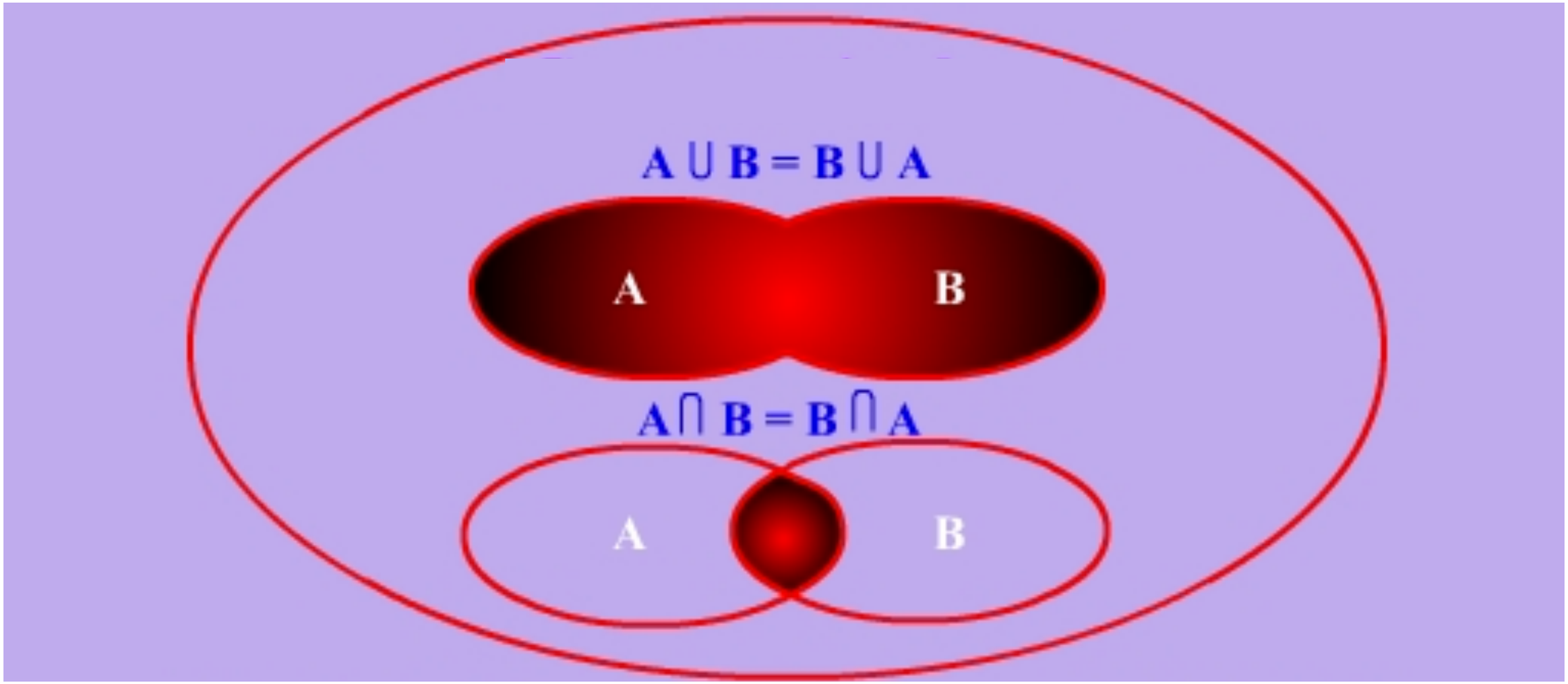
- Let $A_1 = \{x, y\}$, $A_2 = \{1, 2, 3\}$, and $A_3 = \{a, b\}$. Find:
  (a) $A_1 \times A_2$,
  (b) $(A_1 \times A_2) \times A_3$, and

Proof using set identities (Algebraic proof)

Set identities

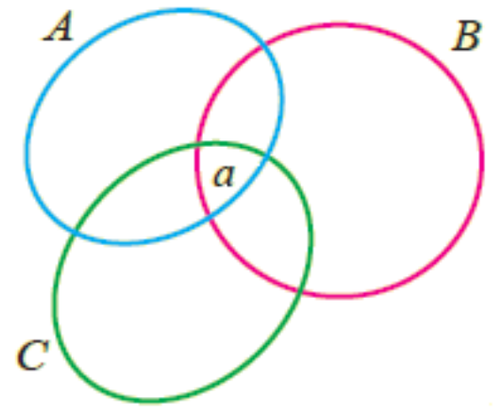
Laws	Formula	Formula
Commutative laws	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive laws	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity laws	$A \cup \phi = A$	$A \cap U = A$
Complement laws	$A \cup A' = U$	$A \cap A' = \phi$
Double comp. law	$(A')' = A$	
Idempotent laws	$A \cup A = A$	$A \cap A = A$
Uni. bound laws	$A \cup U = U$	$A \cap \phi = \phi$
De Morgan's laws	$(A \cup B)' = A' \cap B'$	$(A \cap B)' = A' \cup B'$
Absorption laws	$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$
Complements	$U' = \phi$	$\phi' = U$
Set diff. laws	$A - B = A \cap B'$	

Commutative laws



Associative laws

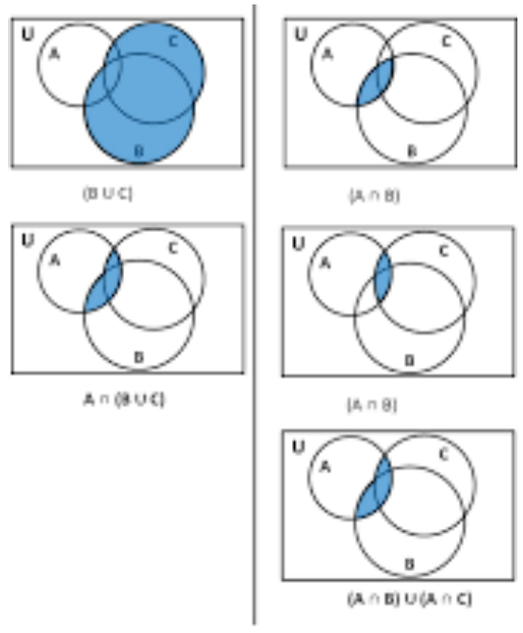
$$(A \cup B) \cup C = A \cup (B \cup C) \qquad (A \cap B) \cap C = A \cap (B \cap C)$$



Distributive laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$



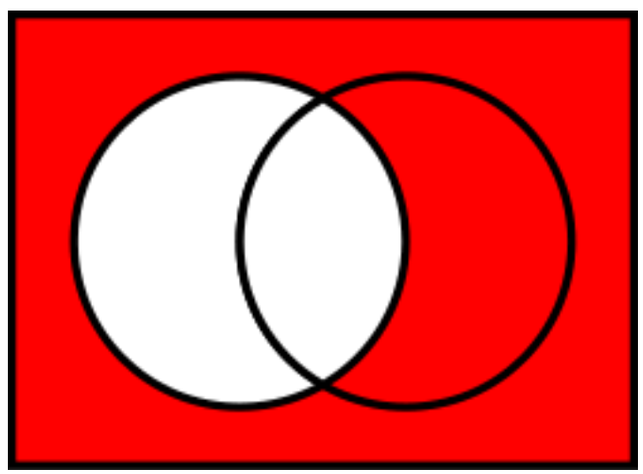
Identity, Idempotent & Universal Bound laws

Identity laws	$A \cup \phi = A$	$A \cap U = A$
Idempotent laws	$A \cup A = A$	$A \cap A = A$
Uni. bound laws	$A \cup U = U$	$A \cap \phi = \phi$

Complement & Double complement laws

$A \cup A' = U$	$A \cap A' = \phi$

$$(A')' = A$$



De Morgen Laws

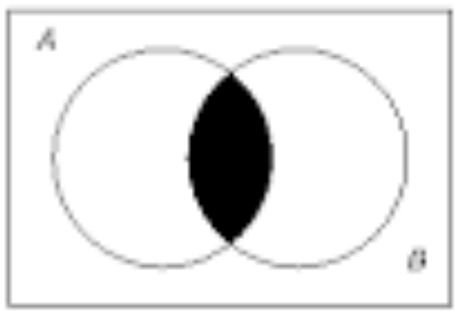


$$(A \cup B)' = A' \cap B' \qquad (A \cap B)' = A' \cup B'$$

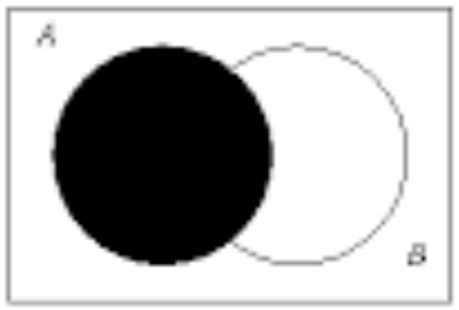
Absorption laws

Absorption laws $A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$

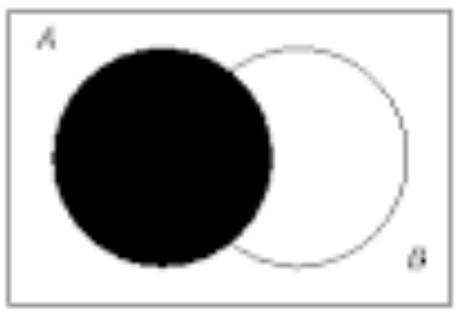
$A \cap B$



A



$A \cup (A \cap B)$



Example

Proposition

- Construct an algebraic proof that for all sets A, B, and C, $(A \cup B) - C = (A - C) \cup (B - C)$

Proof

- $(A \cup B) - C$
  $= (A \cup B) \cap C'$ ($\because$ set difference law)
  $= C' \cap (A \cup B)$ ($\because$ commutative law)
  $= (C' \cap A) \cup (C' \cap B)$ ($\because$ distributive law)
  $= (A \cap C') \cup (B \cap C')$ ($\because$ commutative law)
  $= (A - C) \cup (B - C)$ ($\because$ set difference law)

Exercises

1. Prove that for any sets A and B, $A \cup (A' \cap B) = A \cup B$. [Hint: Use set identities]

Exercises

2. Prove that for any sets A, B and C, $(A - B) - (B - C) = A - B$. [Hint: Use set identities]

Exercises

3. Prove that for any sets A, B, and C, $(A - C) \cap (C - B) = \emptyset$. [Hint: Use set identities]

Review on Sets

1. Membership and Subsets

Basic Rules

- a ∈ A if you can find the exact “a” in A
- Translate $a \notin A$ as ~ (a ∈ A)
- Translate B ⊆ A to “for any b, b∈ B -> b∈ A”
- Emptyset is a subset of any set

Examples

1. $1 \in \{1,\{1\}\}$ ..... 1 is the first element listed in $\{1,\{1\}\}$
2. $1 \notin \{1,\{1\}\}$ ..... because 1 is not a set
3. $\{1\} \in \{1,\{1\}\}$ ..... $\{1\}$ is the second element listed in $\{1,\{1\}\}$

Examples

4. $\{1\} \subseteq \{1, \{1\}\}$ . . . . . make subset $\{1\}$ by selecting 1 from $\{1, \{1\}\}$
5. $\{\{1\}\} \notin \{1, \{1\}\}$ . . . . . because $\{1, \{1\}\}$ contains only 1 and $\{1\}$, and not $\{\{1\}\}$
6. $\{\{1\}\} \subseteq \{1, \{1\}\}$ . . . . . make subset $\{\{1\}\}$ by selecting $\{1\}$ from $\{1, \{1\}\}$

Examples

7. $\mathbb{N} \notin \mathbb{N}$ . . . . . $\mathbb{N}$ is a set (not a number) and $\mathbb{N}$ contains only numbers
8. $\mathbb{N} \subseteq \mathbb{N}$ . . . . . because $X \subseteq X$ for every set X
9. $\emptyset \notin \mathbb{N}$ . . . . . because the set $\mathbb{N}$ contains only numbers and no sets

Examples

10. $\emptyset \subseteq \mathbb{N}$ ..... because $\emptyset$ is a subset of every set
11. $\mathbb{N} \in \{\mathbb{N}\}$ ..... because $\{\mathbb{N}\}$ has just one element, the set $\mathbb{N}$
12. $\mathbb{N} \not\subseteq \{\mathbb{N}\}$ ..... because, for instance, $1 \in \mathbb{N}$ but $1 \notin \{\mathbb{N}\}$

Quiz: True or False

$$\{(1,2),(2,2),(7,1)\} \subseteq \mathbb{N} \times \mathbb{N}$$

2. Powerset

Definition : If A is a set, the **power set** of A is another set, denoted as $\mathscr{P}(A)$ and defined to be the set of all subsets of A. In symbols, $\mathscr{P}(A) = \{X : X \subseteq A\}$.

POWER SETS

- If *S* is the set {a, b, c} then {a,c} is a subset of *S*. There are other subsets of *S*; the complete list is as follows:
  - {} (the empty set or null set)
  - {a}
  - {b}
  - {c}
  - {a, b}
  - {a, c}
  - {b, c}
  - {a, b, c} or S
- So the power set of *S*, written *P*(*S*), is the set containing all the subsets above. Written out this would be the set:
- *P*(*S*) = { {}, {a}, {b}, {c}, {a, b}, {a, c}, {b, c}, {a, b, c} }

Fact

If A is a finite set, then $|\mathscr{P}(A)| = 2^{|A|}$.

|A| refers to # of elements in a set A

Basic Rules

- Know how to get the powerset of N elements, for N>=0.
- Emptyset is always an element of the powerset
- The set itself is also always an element of powerset

Examples

1. $\mathscr{P}(\{0,1,3\}) = \{\, \varnothing, \{0\}, \{1\}, \{3\}, \{0,1\}, \{0,3\}, \{1,3\}, \{0,1,3\} \,\}$
2. $\mathscr{P}(\{1,2\}) = \{\, \varnothing, \{1\}, \{2\}, \{1,2\} \,\}$
3. $\mathscr{P}(\{1\}) = \{\, \varnothing, \{1\} \,\}$

Examples

4. $\mathscr{P}(\emptyset) = \{\emptyset\}$
5. $\mathscr{P}(\{a\}) = \{\emptyset, \{a\}\}$
6. $\mathscr{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$

The following are wrong . Explain why

$$\mathscr{P}(1) = \{\, \emptyset, \{1\} \,\} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$\mathscr{P}\left(\{1, \{1,2\}\}\right) = \{\emptyset, \{1\}, \{1,2\}, \{1, \{1,2\}\}\} \ldots.$$

$$\mathscr{P}\left(\{1, \{1,2\}\}\right) = \{\emptyset, \{\{1\}\}, \{\{1,2\}\}, \{\emptyset, \{1,2\}\}\}$$

Prove set properties using element argument

Facts for element arguments

$A \subseteq B \ \textit{if and only if} \ \forall x, x \in A \rightarrow x \in B$

Facts for element arguments

Let X and Y be subsets of a universal set U and suppose x and y are elements of U.

- $x \in X \cup Y \Leftrightarrow x \in X$ or $x \in Y$
- $x \in X \cap Y \Leftrightarrow x \in X$ and $x \in Y$
- $x \in X - Y \Leftrightarrow x \in X$ and $x \notin Y$

- $x \in X' \Leftrightarrow x \notin X$
- $(x, y) \in X \times Y \Leftrightarrow x \in X$ and $y \in Y$

Element argument — situation 1

Basic method for proving that a set is a subset of another

- Let sets X and Y be given. To prove that $X \subseteq Y$,
  1. suppose that x is a particular but arbitrarily chosen element of X.
  2. show that x is an element of Y.

Element argument — situation 2

Basic method for proving that two sets are equal

- Let sets X and Y be given. To prove that $X = Y$,
  1. Prove that $X \subseteq Y$.
  2. Prove that $Y \subseteq X$.

Element argument — situation 3

Basic method for proving a set equals the empty set

- To prove that a set X is equal to the empty set ϕ, prove that X has no elements.
- To do this, suppose X has an element and derive a contradiction.

Proof by element argument: Example 1

Proposition

- Prove that for all sets A, B, and C
  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof

We need to prove:

1. $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$
2. $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$

Proof by element argument: Example 1

Proof (continued)

Proof that $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$.

Suppose $x \in A \cup (B \cap C)$.

$x \in A$ or $x \in B \cap C$ ($\because$ defn. of union)

- Case 1. $[x \in A.]$

  $x \in A \cup B$ ($\because$ defn. of union)

  $x \in A \cup C$ ($\because$ defn. of union)

  $x \in (A \cup B) \cap (A \cup C)$ ($\because$ defn. of intersection)

- Case 2. $[x \in B \cap C.]$

  $x \in B$ and $x \in C$ ($\because$ defn. of intersection)

  $x \in A \cup B$ ($\because$ defn. of union)

  $x \in A \cup C$ ($\because$ defn. of union)

  $x \in (A \cup B) \cap (A \cup C)$ ($\because$ defn. of intersection)

Proof by element argument: Example 1

Proof (continued)

Proof that $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$.

Suppose $x \in (A \cup B) \cap (A \cup C)$.

$x \in A \cup B$ and $x \in A \cup C$ ($\because$ defn. of intersection)

- Case 1. $[x \in A.]$

  $x \in A \cup (B \cap C)$ ($\because$ defn. of union)

- Case 2. $[x \notin A.]$

  $x \in A$ or $x \in B$ ($\because$ defn. of union)

  $x \in B$ ($\because x \notin A$)

  $x \in A$ or $x \in C$ ($\because$ defn. of union)

  $x \in C$ ($\because x \notin A$)

  $x \in B \cap C$ ($\because$ defn. of intersection)

  $x \in A \cup (B \cap C)$ ($\because$ defn. of union)

Proof by element argument: Example 2

Proposition

- Prove that for all sets A and B, $(A \cup B)' = A' \cap B'$.

Proof

We need to prove:

1. $(A \cup B)' \subseteq A' \cap B'$
2. $A' \cap B' \subseteq (A \cup B)'$

Proof by element argument: Example 2

Proof (continued)

- Proof that $(A \cup B)' \subseteq A' \cap B'$.
  Suppose $x \in (A \cup B)'$.
  $x \notin A \cup B$ ($\because$ defn. of complement)
  It is false that (x is in A or x is in B).
  x is not in A and x is not in B ($\because$ De Morgan's law of logic)
  $x \notin A$ and $x \notin B$
  $x \in A'$ and $x \in B'$ ($\because$ defn. of complement)
  $x \in (A' \cap B')$ ($\because$ defn. of intersection)
  Hence, $(A \cup B)' \subseteq A' \cap B'$ ($\because$ defn. of subset)

Proof by element argument: Example 2

Proof (continued)

- Proof that $A' \cap B' \subseteq (A \cup B)'$.
  Suppose $x \in A' \cap B'$.
  $x \in A'$ and $x \in B'$ ($\because$ defn. of intersection)
  $x \notin A$ and $x \notin B$ ($\because$ defn. of complement)
  x is not in A and x is not in B
  It is false that (x is in A or x is in B)
  ($\because$ De Morgan's law of logic)
  $x \notin A \cup B$
  $x \in (A \cup B)'$ ($\because$ defn. of complement)
  Hence, $A' \cap B' \subseteq (A \cup B)'$ ($\because$ defn. of subset)

Proof by element argument: Example 3

Proposition

- For any sets A and B, if $A \subseteq B$, then
  (a) $A \cap B = A$ and (b) $A \cup B = B$.

Proof

Part (a): We need to prove:

1. $A \cap B \subseteq A$
2. $A \subseteq A \cap B$

Part (b): We need to prove:

1. $A \cup B \subseteq B$
2. $B \subseteq A \cup B$

Proof by element argument: Example 3

Proof (continued)

Part (a).

1. Proof that $A \cap B \subseteq A$.

   $A \cap B \subseteq A$ $(\because$ inclusion of intersection$)$

2. Proof that $A \subseteq A \cap B$.

   Suppose $x \in A$

   $x \in B$ $(\because A \subseteq B)$

   $x \in A$ and $x \in B$

   $x \in A \cap B$ $(\because$ defn. of intersection$)$

Proof by element argument: Example 3

Proof (continued)

Part (b).

1. Proof that $A \cup B \subseteq B$.

   Suppose $x \in A \cup B$

   $x \in A$ or $x \in B$ $(\because$ defn. of union$)$

   If $x \in A$, then $x \in B$ $(\because A \subseteq B)$

   $x \in B$ $(\because$ Modus Ponens and division into cases$)$

2. Proof that $B \subseteq A \cup B$.

   $B \subseteq A \cup B$ $(\because$ inclusion in union$)$

Proof by element argument: Example 4

Proposition

- If E is a set with no elements and A is any set, then $E \subseteq A$.

Proof

Proof by contradiction.

- Suppose there exists a set E with no elements and a set A such that $E \not\subseteq A$.
- $\exists x$ such that $x \in E$ and $x \notin A$ ($\because$ defn. of a subset)
- But there can be no such element since E has no elements.
- Contradiction!
- Hence, if E is a set with no elements and A is any set, then $E \subseteq A$.

Proof by element argument: Example 5

Proposition

- Prove that for any set A, $A \cap \phi = \phi$

Proof

Proof by contradiction.

- Suppose there is an element x such that $x \in A \cap \phi$
- $x \in A$ and $x \in \phi$ ($\because$ defn. of intersection)
- $x \in \phi$
- Impossible because ϕ cannot have any elements
- Hence, the supposition is incorrect.
- So, $A \cap \phi = \phi$